

TAK Server - AD/LDAP

You should have a TAK Server fully stood up to proceed.

TAK Serve - LDAP Integration

- Update your server:
 - **sudo dnf update -y**
- Install OpenLDAP Packages:
 - **sudo dnf config-manager --set-enabled plus**
 - **sudo dnf install openldap openldap-servers openldap-clients -y**
- Enable and start the LDAP service:
 - **sudo systemctl enable slapd --now**

```
[takadmin@localhost ~]$ sudo systemctl enable slapd --now
Created symlink /etc/systemd/system/openldap.service → /usr/lib/systemd/system/slapd.service.
Created symlink /etc/systemd/system/multi-user.target.wants/slapd.service → /usr/lib/systemd/system/slapd.service.
```

TAK Server - LDAP Integration

- Verify the service is running:
 - **sudo systemctl status slapd**

```
[takadmin@localhost ~]$ sudo systemctl status slapd
● slapd.service - OpenLDAP Server Daemon
   Loaded: loaded (/usr/lib/systemd/system/slapd.service; enabled; vendor preset: disabled)
   Active: active (running) since Sat 2024-08-10 10:46:15 EDT; 8s ago
     Docs: man:slapd
           man:slapd-config
           man:slapd-hdb
           man:slapd-mdb
           file:///usr/share/doc/openldap-servers/guide.html
   Process: 84718 ExecStart=/usr/sbin/slapd -u ldap -h ldap:/// ldaps:/// ldapi:/// (code=exited, status=0/SUCCESS)
   Process: 84704 ExecStartPre=/usr/libexec/openldap/check-config.sh (code=exited, status=0/SUCCESS)
  Main PID: 84719 (slapd)
    Tasks: 2 (limit: 61771)
   Memory: 3.0M
   CGroup: /system.slice/slapd.service
           └─84719 /usr/sbin/slapd -u ldap -h ldap:/// ldaps:/// ldapi:///

Aug 10 10:46:14 localhost.localdomain systemd[1]: Starting OpenLDAP Server Daemon...
Aug 10 10:46:14 localhost.localdomain runuser[84707]: pam_unix(runuser:session): session opened for user ldap by (uid=0)
Aug 10 10:46:14 localhost.localdomain runuser[84707]: pam_unix(runuser:session): session closed for user ldap
Aug 10 10:46:15 localhost.localdomain slapd[84718]: @(#) $OpenLDAP: slapd 2.4.46 (May  1 2024 00:39:23) $
                        mockbuild@ord1-prod-x86build005.svc.aws.rockylinux.org:/bui
Aug 10 10:46:15 localhost.localdomain slapd[84719]: slapd starting
Aug 10 10:46:15 localhost.localdomain systemd[1]: Started OpenLDAP Server Daemon.
```

TAK Server - LDAP Integration

- Configure OpenLDAP
 - Generate a hashed password for the LDAP administrator:

- **slappasswd**

- Enter in a password of your choice: password
 - Copy and save the hashed password.

```
[takadmin@localhost ~]$ slappasswd
New password:
Re-enter new password:
{SSHA}78Pp9CLvQDGmaCCGyYnHXZ17CS2Q+Niu
```

- Check the databases backend before modifying:
 - **sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config dn**
 - The last entry is the important one we need to note.

```
[takadmin@localhost ~]$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config dn
dn: cn=config

dn: cn=schema,cn=config
dn: cn={0}core,cn=schema,cn=config
dn: olcDatabase={-1}frontend,cn=config
dn: olcDatabase={0}config,cn=config
dn: olcDatabase={1}monitor,cn=config
dn: olcDatabase={2}mdb,cn=config
```

TAK Server - LDAP Integration

- Create a LDIF file (“db.ldif”) to configure the domain and admin credentials:
 - Create the file: **vim db.ldif**
 - Add the following to the file:
 - olcRootPW: add your hashed password
 - To save and exit:
 - Press “esc” key
 - Type: :wq!
 - Press enter
 - Apply the configuration:
 - **sudo ldapmodify -Y EXTERNAL -H ldapi:/// -f db.ldif**

```
dn: olcDatabase={2}mdb,cn=config
changetype: modify
replace: olcSuffix
olcSuffix: dc=example,dc=com

dn: olcDatabase={2}mdb,cn=config
changetype: modify
replace: olcRootDN
olcRootDN: cn=admin,dc=example,dc=com

dn: olcDatabase={2}mdb,cn=config
changetype: modify
replace: olcRootPW
olcRootPW: {SSHA}78Pp9CLvQDGmaCCGyYnHXZ17CS2Q+Niu
```

TAK Server - LDAP Integration

- Add the necessary schemas to the LDAP directory:
 - **sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/cosine.ldif**
 - **sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/nis.ldif**
 - **sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/inetorgperson.ldif**

```
[takadmin@localhost ~]$ sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/cosine.ldif
SASL/EXTERNAL authentication started
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
SASL SSF: 0
adding new entry "cn=cosine,cn=schema,cn=config"

[takadmin@localhost ~]$ sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/nis.ldif
SASL/EXTERNAL authentication started
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
SASL SSF: 0
adding new entry "cn=nis,cn=schema,cn=config"

[takadmin@localhost ~]$ sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/inetorgperson.ldif
SASL/EXTERNAL authentication started
SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
SASL SSF: 0
adding new entry "cn=inetorgperson,cn=schema,cn=config"
```

TAK Server - LDAP Integration

- To verify if the schemas were applied run the following:
 - **sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=schema,cn=config dn**

```
[takadmin@localhost ~]$ sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=schema,cn=config dn
dn: cn=schema,cn=config

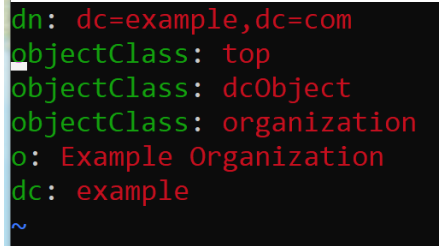
dn: cn={0}core,cn=schema,cn=config

dn: cn={1}cosine,cn=schema,cn=config

dn: cn={2}nis,cn=schema,cn=config

dn: cn={3}inetorgperson,cn=schema,cn=config
```

TAK Server - LDAP Integration

- To check existing OUs (Organizational Units):
 - **sudo ldapsearch -x -LLL -b dc=example,dc=com “(objectClass=organizationalUnit)” dn**
- To create a base DN, create a LDIF file (“add-base.ldif”)
 - **vim add-base.ldif**
 - Add the following to the file:
 - Add the changes:
 - **sudo ldapadd -x -D “cn=admin,dc=example,dc=com” -W -f add-base.ldif**

```
[takadmin@localhost ~]$ sudo ldapadd -x -D "cn=admin,dc=example,dc=com" -W -f add-base.ldif
Enter LDAP Password:
adding new entry "dc=example,dc=com"
```


TAK Server - LDAP Integration

- To create a Organizational Unit (OU):

- Create a LDIF file (“ou.ldif”)
 - **vim ou.ldif**
 - Add the following to the file: →
- Apply the configuration:
 - **ldapadd -x -D “cn=admin,dc=example,dc=com” -W -f ou.ldif**

```
dn: ou=People,dc=example,dc=com
objectClass: organizationalUnit
ou: People
-
dn: ou=Groups,dc=example,dc=com
objectClass: organizationalUnit
ou: Groups
~
```

```
[takadmin@localhost ~]$ [takadmin@localhost ~]$ ldapadd -x -D "cn=admin,dc=example,dc=com" -W -f ou.ldif
Enter LDAP Password:
adding new entry "ou=People,dc=example,dc=com"

adding new entry "ou=Groups,dc=example,dc=com"
```

TAK Server - LDAP Integration - Managing Users and Groups

- Add a user:
 - Create a new LDIF file (“new_user.ldif”)
 - **vim new_user.ldif**
 - Add the following to the file: →
 - Apply the changes:
 - **sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f new_user.ldif**
- Verify the changes:
 - **sudo ldapsearch -x -LLL -b dc=example,dc=com “(uid=jdoe)”**

```
dn: uid=jdoe,ou=People,dc=example,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: top
cn: John Doe
sn: Doe
uidNumber: 1001
gidNumber: 1001
homeDirectory: /home/jdoe
loginShell: /bin/bash
mail: jdoe@example.com
userPassword: {SSHA}eIW6vVR8aV8M6q6y007vrUZBcUYS2PJ1
```

```
[takadmin@localhost ~]$ [takadmin@localhost ~]$ sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f new_user.ldif
Enter LDAP Password:
adding new entry "uid=jdoe,ou=People,dc=example,dc=com"
```

TAK Server - LDAP Integration - Managing Users and Groups

- Create a new Group:

- Create a new LDIF file (“new_group.ldif”)

- **vim new_group.ldif**

- Add the following to the file: →

```
dn: cn=Devolpers,ou=Groups,dc=example,dc=com
objectClass: top
objectClass: posixGroup
cn: developers
gidNumber: 1001
```

- Apply the changes:

- **sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f new_group.ldif**

```
[takadmin@localhost ~]$ sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f new_group.ldif
Enter LDAP Password:
adding new entry "cn=Devolpers,ou=Groups,dc=example,dc=com"
```

- Verify the changes:

- **sudo ldapsearch -x -LLL -b dc=example,dc=com “(cn=Devolpers)”**

TAK Server - LDAP Integration - Modify Users and Groups

- To modify a user:

- Create a LDIF file (“modify_user.ldif”)
 - **vim modify_user.ldif**
 - Add the following to the file: →
 - This example changes the users password.

```
dn: uid=jdoe,ou=People,dc=example,dc=com
changetype: modify
replace: userPassword
userPassword: SSHA password
```

- Apply the changes:
 - **sudo ldapmodify -x -D cn=admin,dc=example,dc=com -W -f modify_user.ldif**
- Verify the changes;
 - **sudo ldapsearch -x -LLL -b dc=example,dc=com “(uid=jdoe)”**

TAK Server - LDAP Integration - Modify Users and Groups

- To modify a group:
 - Create a new LDIF file (“modify_group.ldif”)
 - **vim modify_group.ldif**
 - Add the following to the file: →
 - This example adds a user to a group.
 - Apply the changes:
 - **sudo ldapmodify -x -D cn=admin,dc=example,dc=com -W -f modify_group.ldif**
 - Verify the changes:
 - **sudo ldapsearch -x -LLL -b dc=example,dc=com “(cn=Developers)”**

```
dn: cn=Developers,ou=Groups,dc=example,dc=com_
changetype: modify
add: memberUid
memberUid: jdoe
~
```

TAK Server - LDAP Integration - Managing Users and Groups

- To delete a user:
 - Create a LDIF file (“delete_user.ldif”)
 - **vim delete_user.ldif**
 - Add the following to the file: →
 - Apply the changes:
 - **sudo ldapdelete -x -D cn=admin,dc=example,dc=com -W uid=jdoe,ou=People,dc=example,dc=com**
 - Verify the changes:
 - **sudo ldapsearch -x -LLL -b dc=example,dc=com “(uid=jdoe)”**
 - There should be an empty response since the user was deleted.

```
dn: uid=jdoe,ou=People,dc=example,dc=com
changetype: delete_
```

TAK Server - LDAP Integration - Managing Users and Groups

- To Delete a group:
 - Create a new LDIF file (“delete_group.ldif”)
 - **vim delete_group.ldif**
 - Add the following to the file: →
 - Apply the changes:
 - **sudo ldapdelete -x -D cn=admin,dc=example,dc=com -W cn=Developers,ou=Groups,dc=example,dc=com**
 - Verify the changes:
 - **sudo ldapsearch -x -LLL -b dc=example,dc=com “(cn=Developers)”**
 - You should not get any response since the group was deleted.

```
dn: cn=Developers,ou=Groups,dc=example,dc=com
changetype: delete
~
```

TAK Server -LDAP Integration - Connecting to TAK Server

- Verify the default port needed for LDAP is configured:
 - **sudo firewall-cmd --list-all**
 - If port 389/tcp is not enabled do the following:
 - **sudo firewall-cmd --zone=public --add-port 389/tcp --permanent**
 - Reload the firewall:
 - **sudo firewall-cmd --reload**
 - Verify the port was added.
 - **sudo firewall-cmd --list-all**

```
[takadmin@localhost ~]$ [takadmin@localhost ~]$  
[sudo] password for takadmin:  
public (active)  
  target: default  
  icmp-block-inversion: no  
  interfaces: enp0s3 enp0s8  
  sources:  
  services: cockpit dhcpv6-client ssh  
  ports: 8089/tcp 8443/tcp 8446/tcp 389/tcp  
  protocols:  
  forward: no  
  masquerade: no  
  forward-ports:  
  source-ports:  
  icmp-blocks:  
  rich rules:  
[takadmin@localhost ~]$
```


TAK Server -LDAP Integration - Connecting to TAK Server

- To configure the TAK Server:
 - Enter the tak user:
 - **sudo su tak**
 - Open the CoreConfig file with a text editor:
 - **vim /opt/tak/CoreConfig.xml**
 - Add the following:
 - `<auth default="ldap" x509groups="true" x509addAnonymous="false">`
 - `<ldap url="ldap://192.168.1.119:389"`
 - `userstring="uid={username},ou=People,dc=example,dc=com" updateinterval="60"`
 - `groupNameExtractorRegex="CN=(.*?)(?:,|$)" style="DS"`
 - `serviceAccountDN="cn=admin,dc=example,dc=com"`
 - `serviceAccountCredential="password" groupObjectClass="groupOfNames"`
 - `userObjectClass="inetOrgPerson" groupBaseRDN="dc=example,dc=com"`
 - `userBaseRDN="ou=People,dc=example,dc=com"/>`
 - `</auth>`

TAK Server -LDAP Integration - Connecting to TAK Server

- The additions to the CoreConfig.xml file should look as follows:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<Configuration xmlns="http://bbn.com/marti/xml/config">
  <network multicastTtl="5" serverId="239fd331666342a5acbf98192edb94de" version="5.2-RELEASE-16-HEAD">
    <input _name="stdssl" protocol="tls" port="8089" coreVersion="2"/>
    <connector port="8443" _name="https"/>
    <connector port="8444" useFederationTruststore="true" _name="fed_https"/>
    <connector port="8446" clientAuth="false" _name="cert_https"/>
    <announce/>
  </network>
  <auth default="ldap" x509groups="true" x509addAnonymous="false">
    <ldap url="ldap://192.168.1.119:389" userstring="uid={username},ou=People,dc=example,dc=com" updateinterval="60" groupNameExtractorRegex="CN=(.*?)(?:,|$)" style="DS" serviceAccountDN="cn=admin,dc=example,dc=com" serviceAccountCredential="password" groupObjectClass="groupOfNames" userObjectClass="inetOrgPerson" groupBaseRDN="dc=example,dc=com" userBaseRDN="ou=People,dc=example,dc=com"/>
    <File/>
  </auth>
  <submission ignoreStaleMessages="false" validateXml="false"/>
  <subscription reloadPersistent="false"/>
  <repository enable="true" numDbConnections="200" primaryKeyBatchSize="500" insertionBatchSize="500">
    <connection url="jdbc:postgresql://127.0.0.1:5432/cot" username="martiuser" password="wZUpiKubPt6Agqcy"/>
  </repository>
  <repeater enable="true" periodMillis="3000" staleDelayMillis="15000">
    <repeatableType initiate-test="/event/detail/emergency[@type='911 Alert']" cancel-test="/event/detail/emergency[@cancel='true']" _name="911"/>
    <repeatableType initiate-test="/event/detail/emergency[@type='Ring The Bell']" cancel-test="/event/detail/emergency[@cancel='true']" _name="RingTheBell"/>
    <repeatableType initiate-test="/event/detail/emergency[@type='Geo-fence Breached']" cancel-test="/event/detail/emergency[@cancel='true']" _name="GeoFenceBreach"/>
    <repeatableType initiate-test="/event/detail/emergency[@type='Troops In Contact']" cancel-test="/event/detail/emergency[@cancel='true']" _name="TroopsInContact"/>
  </repeater>
  <filter>
    <thumbnail/>
    <urladd host="http://127.0.0.1:8080"/>
    <flowtag enable="true" text=""/>
    <streamingbroker enable="true"/>
    <scrubber enable="false" action="overwrite"/>
    <qos>
      <deliveryRateLimiter enabled="true">
        <rateLimitRule clientThresholdCount="500" reportingRateLimitSeconds="200"/>
        <rateLimitRule clientThresholdCount="1000" reportingRateLimitSeconds="300"/>
      </deliveryRateLimiter>
    </qos>
  </filter>
</Configuration>
```

TAK Server -LDAP Integration - Connecting to TAK Server

- Once the changes have been made restart the TAK Server:
 - **sudo systemctl restart takserver**
 - You can monitor the system to know when the server is back online with:
 - **tail -f /opt/tak/logs/takserver-messaging.log**
- When the server is back online you can verify the LDAP service is running as expected by trying to logging with the user credentials created through the LDAP service at the following:
 - `https://<IP address>:8446`
 - If you are able to login then you have successfully integrated the LDAP service.

TAK Server -LDAP Query

- To query users and groups from your database:
 - Use yum as it provides commands and options to search for the parent package:
 - **sudo yum whatprovides ldapsearch**
 - Install the openldap-clients
 - **sudo yum install openldap-clients -y**
 - Search for the syntax information:
 - **ldapsearch -LLL -H ldap://19.168.1.119:389 -b 'dc=example,dc=com' -x -D 'example/tak.service' -W '(sAMAccountName=<user>)'**

TAK Server - LDAP Group Prefix

- IN and OUT Groups with LDAP:
 - To apply the in/out filters to a group with LDAP you need to create an additional group with the same naming but with either “_READ” or “_WRITE” at the end of the name.

- _READ: out group
- _WRITE: in group

Name	Type	Description
 tak_GroupA	Security Group - Global	
 tak_GroupA_READ	Security Group - Global	OUT Group
 tak_GroupA_WRITE	Security Group - Global	IN Group
 tak_GroupB	Security Group - Global	
 tak_GroupB_READ	Security Group - Global	OUT Group
 tak_GroupB_WRITE	Security Group - Global	IN Group

- If you somehow forget to add a user to a group, they will default to the __ANON__ group when they authenticate to the TAK Server.

Resources

- <https://www.golinuxcloud.com/install-configure-openldap-rocky-linux-8/>

TAK Server - Ports

- Use the following to print the PID on a specific port:
 - **fuser 8080/tcp**
 - **fuser 8080/udp**
- Use the following to kill a process on a port:
 - **fuser -k 8080/tcp**
 - **fuser -k 8080/udp**
- https://docs.redhat.com/en/documentation/red_hat_directory_server/12/html/installing_red_hat_directory_server/changing-the-ldap-and-ldaps-port-numbers_installing-rhds#changing-the-port-numbers-using-the-command-line_changing-the-ldap-and-ldaps-port-numbers